

User Acceptance Testing (UAT) Summary Report

Date of Execution: June 11, 2026

Test Status: 100% Passed

Executive Summary

This report document outlines the final results of the User Acceptance Testing (UAT) phase. All executed scenarios have met their expected results with zero critical, major, or minor defects recorded.

The application demonstrated excellent state management, precise client-side mathematical processing, seamless asynchronous request chunking for extended historical periods, and graceful error handling under simulated network/data constraints.

Testing Summary Dashboard

Scenario ID	Scenario Description	Expected Outcome	Status
UAT-01	Configuration and Currency Analysis Generation	Request chunking & data merging (>93 days)	PASSED
UAT-02	Statistical Measures & Session Trends Evaluation	Flawless client-side math execution (No NaN/Infinity)	PASSED
UAT-03	Change Distribution Analysis & Timeframe Toggling	Dynamic Recharts histogram rendering	PASSED
UAT-04	Exporting Analysis Results and Visualizations	Successful Blob/Canvas download (CSV, JSON, Images)	PASSED
UAT-05	Alternative Flows and Error Handling	Clean user alerts for API timeouts & non-trading days	PASSED

1. UAT-01: Configuration and Currency Analysis Generation

- **Objective:** Verify currency parameter selection and request chunking mechanism for periods exceeding 93 days.

- **Execution Notes:** Tested with a 1-year historical timeframe. The client application successfully split the request into sequential chunks, managed asynchronous delivery, and merged the results into a single dataset.

2. UAT-02: Statistical Measures & Session Trends Evaluation

- **Objective:** Validate client-side processing of day-to-day session changes and statistical accuracy.

- **Execution Notes:** The browser-based engine handled all calculations locally without server dependencies. Growth, decline, and stable sessions matched the aggregated data sets precisely.

- **Data Integrity:** Median, Mode, Standard Deviation, and Coefficient of Variation were properly formatted. No NaN or Infinity errors occurred.

3. UAT-03: Change Distribution Analysis and Timeframe Toggling

- **Objective:** Verify calculation of session-to-session differences and dynamic histogram updates.

- **Execution Notes:** Toggling between 'Month' and 'Quarter' views triggered instant chart updates. The Recharts-powered frequency histogram and the adjacent Frequency Results Table remained perfectly synchronized.

4. UAT-04: Exporting Analysis Results and Visualizations

- **Objective:** Verify data parsing and local file generation features.

- **Execution Notes:** * **CSV & JSON:** Downloaded files contain well-structured raw data and aggregated statistics.

- **PNG/JPEG:** The canvas utility accurately converted the Recharts SVG elements into high-resolution local image files via temporary Blob URLs.

5. UAT-05: Alternative Flows and Error Handling

- **Objective:** Evaluate application resilience during backend API failure or empty data windows.

- **Execution Notes:** * *Test A (API Outage):* Simulated timeouts were caught gracefully. The application showed a helpful notification instead of crashing.

- *Test B (Non-Trading Windows):* Selecting a weekend/holiday window properly informed the user that no trading data was available for that period.